

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA1613 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 30%

QUESTIONS: 3

Total Marks:30

Annexure C

Case study

<i>Criteria</i>	Understandin g of Case (2.5)	Application of Concepts (2.5)	Analysi s (2)	Solution Design (2)	Clarity (1)	<i>Total(10)</i>
<i>Weightag e</i>	25%	25%	20%	20%	10%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						

Code of Conduct:

*I 192311088 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:P.Siva Kumar

RUBRICS FOR CALCULATION:

Criteria	Weightag e	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
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Understanding of Case	25	Thorough understanding ; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ANNEXURE A

Case Study:

1. A shepherd boy gets bored tending the town's flock. To have some fun he cries out, "Wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them.

One night, the shepherd boy sees a real wolf approaching the flock and calls out, "Wolf!" The villagers refuse to be fooled again and stay in their houses. The hungry wolf turns the flock into lamb chops. The town goes hungry. Panic ensues. Identify TP,FP,FN,TN

2. Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat in 50 of the 70 no cat images. Identify TP,FP,FN,TN and Calculate the Precision and Recall and F1 Score .

3. An email service provider wants to classify emails as **Spam or Not Spam** using features:

- Contains "Offer" (Yes/No)

- Contains “Free” (Yes/No)
- Email Length (Short/Long)

Sample Dataset

Offer Free Length			Class
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

Questions

1. Calculate prior probabilities:
 - $P(\text{Spam})$, $P(\text{Not Spam})$
2. Compute conditional probabilities for each feature.
3. Using **Naïve Bayes**, classify:
 - (Offer=Yes, Free=No, Length=Short)
4. Compare results using **WEKA (Naïve Bayes classifier)**.
5. Discuss the **independence assumption** in Naïve Bayes

1. Confusion Matrix for the Shepherd Boy and Wolf Problem

We can consider:

- **Positive (+)** = A wolf is actually present / a "Wolf!" warning is given.

- **Negative (-)** = No wolf is present / no "Wolf!" warning is given.
- **Actual condition** = Whether a wolf is really present.
- **Prediction** = Whether the boy gives a "Wolf!" warning.

Wolf Present (Actual +) Wolf Absent (Actual -)

"Wolf!" Warning (+)	TP	FP
No "Wolf!" Warning (-)	FN	TN

1. True Positive (TP):

A **real wolf** is **approaching the flock**, and the shepherd boy correctly cries **"Wolf!"**.

- Actual condition: Wolf is present.
- Prediction/warning: Wolf is present.
- The warning is correct.

Therefore:

TP = Real wolf + Warning given

2. False Positive (FP):

There is **no wolf**, but the shepherd boy cries **"Wolf!"** as a joke.

- Actual condition: No wolf.
- Prediction/warning: Wolf is present.
- The warning is incorrect.

Therefore:

FP = No wolf + False warning

This happens when the villagers run to protect the flock even though there is no danger.

3. False Negative (FN)

A **real wolf** is **present**, but the warning is **not recognized or acted upon**.

In the story, the boy actually calls "Wolf!" when the real wolf appears, but the villagers ignore the warning because of his previous false alarms.

For the classification interpretation, this represents a **missed detection**:

- Actual condition: Wolf is present.
- Effective prediction/action: No warning response.
- The real danger is missed.

Therefore:

FN = Real wolf + Danger missed

4. True Negative (TN):

There is **no wolf**, and the boy **does not give a false "Wolf!" warning**.

- Actual condition: No wolf.
- Prediction: No wolf.
- The decision is correct.

Therefore:

TN = No wolf + No warning

Final Answer:

Term Meaning in this problem

TP Wolf is present and the warning correctly identifies it

FP No wolf is present, but the boy falsely cries "Wolf!"

FN Wolf is present, but the danger is missed/not effectively detected

TN No wolf is present and no false warning is given

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter
Choose: **None** Apply Stop

Current relation
Relation: wolf_prediction
Instances: 2
Attributes: 2
Sum of weights: 2

Selected attribute
Name: Actual
Missing: 0 (0%)
Distinct: 2
Type: Nominal
Unique: 2 (100%)

Attributes
All None Invert Pattern

No.	Label	Count	Weight
1	Wolf	1	1
2	NoWolf	1	1

Class: Predicted (Nom) Visualize All

Remove

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier
Choose: **J48 -C 0.25 -M 2**

Test options
☐ Use training set
☐ Supplied test set Set...
☒ Cross-validation Folds: **10**
☐ Percentage split % **66**
 More options...

(Nom) Predicted
Start Stop

Result list (right-click for options)
17:50:38 - trees.J48

Classifier output

```

=== Run information ===

Scheme:      weka.classifiers.trees.J48 -C 0.25 -M 2
Relation:    wolf_prediction
Instances:   2
Attributes:  2
              Actual
              Predicted
Test mode:   10-fold cross-validation

=== Classifier model (full training set) ===

J48 pruned tree
-----
: Wolf (2.0/1.0)

Number of Leaves :    1

Size of the tree :    1

Time taken to build model: 0 seconds
  
```

Status
Problem evaluating classifier Log x 0

2. Confusion Matrix, Precision, Recall and F1 Score

Given:

- Total images = 100
- Images with cats = 30
- Images without cats = 70
- Model correctly predicts a cat in **25 of the 30 cat images**
- Model correctly predicts no cat in **50 of the 70 non-cat images**

Step 1: Identify TP, FN, TN, and FP:

True Positive (TP):

The image actually contains a cat, and the model correctly predicts a cat.

TP = 25

False Negative (FN):

The image actually contains a cat, but the model predicts no cat.

There are 30 cat images, and 25 were correctly identified:

FN = 30 – 25 = 5

True Negative (TN):

The image does not contain a cat, and the model correctly predicts no cat.

TN = 50

False Positive (FP):

The image does not contain a cat, but the model predicts that there is a cat.

There are 70 non-cat images, and 50 were correctly identified:

FP = 70 – 50 = 20

Confusion Matrix:

	Actual Cat	Actual No Cat
Predicted Cat	TP = 25	FP = 20
Predicted No Cat	FN = 5	TN = 50

So:

- **TP = 25**
- **FP = 20**
- **FN = 5**
- **TN = 50**

Step 2: Calculate Precision:

Precision tells us **how many of the images predicted as cats were actually cats.**

Formula:

Formula:

$$Precision = \frac{TP}{TP + FP}$$

Substitute the values:

$$Precision = \frac{25}{25 + 20}$$

$$Precision = \frac{25}{45}$$

$$Precision = 0.556$$

or

$$Precision = 55.56\%$$

Step 3: Calculate Recall:

Recall tells us how many of the actual cat images were correctly detected.

Formula:

$$Recall = \frac{TP}{TP + FN}$$

$$Recall = \frac{25}{25 + 5}$$

$$Recall = \frac{25}{30}$$

$$Recall = 0.8333$$

or

$$Recall = 83.33\%$$

Step 4: Calculate F1 Score:

F1 Score is the harmonic mean of Precision and Recall.

Formula:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Substitute:

$$F1 = \frac{2 \times 0.556 \times 0.8333}{0.556 + 0.8333}$$

$$F1 \approx 0.667$$

Therefore:

$$F1Score \approx 0.667$$

or

$$F1Score \approx 66.67\%$$

Final Answer:

Measure Value

TP 25

FP 20

FN 5

TN 50

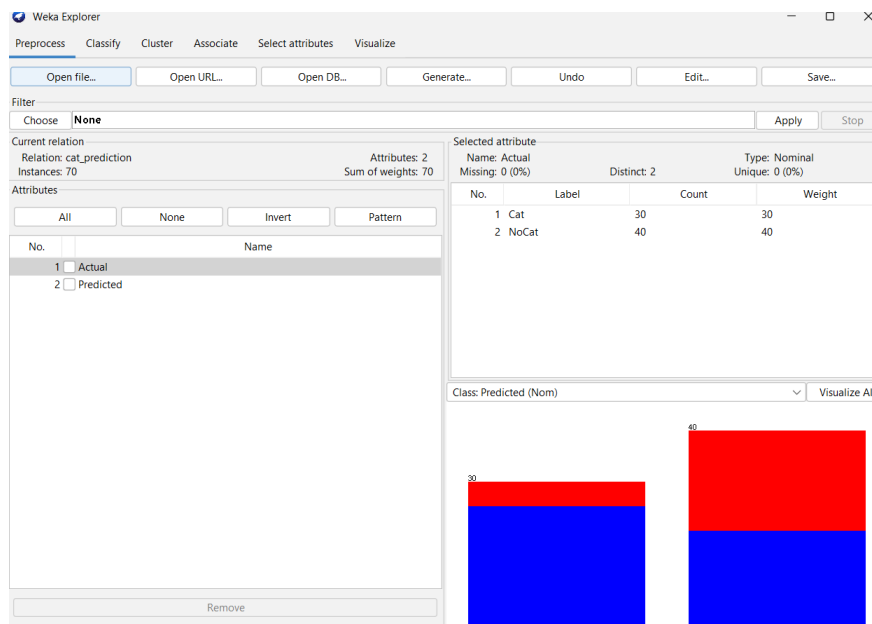
Precision 55.56%

Recall 83.33%

F1 Score 66.67%

Conclusion:

The model has a **high recall (83.33%)**, meaning it detects most of the actual cat images, but its **precision is lower (55.56%)**, meaning many of its cat predictions are incorrect.



Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

ChooseNaiveBayes

Test options

Use training set

Supplied test set

Cross-validation

Folds10

Percentage split

%66

More options...

(Nom) Predicted

StartStop

Result list (right-click for options)

17:50:38 - trees.J48

17:53:36 - bayes.NaiveBayes

Classifier output

Cat	26.0	6.0
NoCat	21.0	21.0
[total]	47.0	27.0

Time taken to build model: 0 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	40	57.1429 %
Incorrectly Classified Instances	30	42.8571 %
Kappa statistic	-0.0244	
Mean absolute error	0.4172	
Root mean squared error	0.4571	
Relative absolute error	90.3871 %	
Root relative squared error	95.148 %	
Total Number of Instances	70	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.778	0.800	0.636	0.778	0.700	-0.026	0.578	0.692	Cat
	0.200	0.222	0.333	0.200	0.250	-0.026	0.578	0.398	NoCat
Weighted Avg.	0.571	0.594	0.528	0.571	0.539	-0.026	0.578	0.587	

=== Confusion Matrix ===

a b <-- classified as

35	10	a = Cat
20	5	b = NoCat

Status

OK

Log

x 0

3. Naïve Bayes Classification – Spam Email

Given Dataset

Offer Free Length Class

Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

Total emails = **6**

- Spam = **3**
- Not Spam = **3**

1. Calculate Prior Probabilities:

The formula is:

$$P(Class) = \frac{\text{Number of emails in Class}}{\text{Total emails}}$$

Probability of Spam

$$P(Spam) = \frac{3}{6} = 0.5$$

Probability of Not Spam

$$P(NotSpam) = \frac{3}{6} = 0.5$$

Therefore:

$$P(Spam) = 0.5$$

$$P(NotSpam) = 0.5$$

2. Calculate Conditional Probabilities

We need to calculate the probability of each feature value given the class.

A. Feature: Offer

For Spam

There are 3 Spam emails:

- Offer = Yes $\rightarrow 2$
- Offer = No $\rightarrow 1$

Therefore:

$$P(Offer = Yes|Spam) = \frac{2}{3} = 0.667$$

$$P(Offer = No|Spam) = \frac{1}{3} = 0.333$$

For Not Spam

There are 3 Not Spam emails:

- Offer = Yes $\rightarrow 1$
- Offer = No $\rightarrow 2$

Therefore:

$$P(\text{Offer} = \text{Yes} | \text{NotSpam}) = \frac{1}{3} = 0.333$$

$$P(\text{Offer} = \text{No} | \text{NotSpam}) = \frac{2}{3} = 0.667$$

B. Feature: Free

For Spam:

Among the 3 Spam emails:

- Free = Yes $\rightarrow 2$
- Free = No $\rightarrow 1$

Therefore:

$$P(\text{Free} = \text{Yes} | \text{Spam}) = \frac{2}{3} = 0.667$$

$$P(\text{Free} = \text{No} | \text{Spam}) = \frac{1}{3} = 0.333$$

For Not Spam:

Among the 3 Not Spam emails:

- Free = Yes $\rightarrow 0$
- Free = No $\rightarrow 3$

Therefore:

$$P(\text{Free} = \text{Yes} | \text{NotSpam}) = \frac{0}{3} = 0$$

$$P(\text{Free} = \text{No} | \text{NotSpam}) = \frac{3}{3} = 1$$

C. Feature: Length

For Spam

Among the 3 Spam emails:

- Short $\rightarrow 2$
- Long $\rightarrow 1$

Therefore:

$$P(\text{Length} = \text{Short} | \text{Spam}) = \frac{2}{3} = 0.667$$

$$P(\text{Length} = \text{Long} | \text{Spam}) = \frac{1}{3} = 0.333$$

For Not Spam

Among the 3 Not Spam emails:

- Short $\rightarrow 1$
- Long $\rightarrow 2$

Therefore:

$$P(\text{Length} = \text{Short} | \text{Not Spam}) = \frac{1}{3} = 0.333$$

$$P(\text{Length} = \text{Long} | \text{Not Spam}) = \frac{2}{3} = 0.667$$

Conditional Probability Table:

Feature	Spam	Not Spam
P(Offer=Yes)	2/3	1/3
P(Offer=No)	1/3	2/3
P(Free=Yes)	2/3	0
P(Free=No)	1/3	1
P(Length=Short)	2/3	1/3
P(Length=Long)	1/3	2/3

3. Classify (Offer=Yes, Free=No, Length=Short)

We need to calculate the probability that the email is **Spam** and **Not Spam**.

The Naïve Bayes formula is:

$$P(Class|Features) \propto P(Class) \times P(Offer|Class) \times P(Free|Class) \times P(Length|Class)$$

Probability of Spam

Given:

- $P(Spam) = 1/2$
- $P(Offer = Yes|Spam) = 2/3$
- $P(Free = No|Spam) = 1/3$
- $P(Length = Short|Spam) = 2/3$

Therefore:

$$\begin{aligned} P(Spam|X) &\propto \frac{1}{2} \times \frac{2}{3} \times \frac{1}{3} \times \frac{2}{3} \\ &= \frac{4}{54} \\ &= \boxed{0.0741} \end{aligned}$$

Probability of Not Spam:

Given:

- $P(NotSpam) = 1/2$
- $P(Offer = Yes|NotSpam) = 1/3$
- $P(Free = No|NotSpam) = 1$
- $P(Length = Short|NotSpam) = 1/3$

Therefore:

$$\begin{aligned}P(NotSpam|X) &\propto \frac{1}{2} \times \frac{1}{3} \times 1 \times \frac{1}{3} \\&= \frac{1}{18} \\&= \boxed{0.0556}\end{aligned}$$

Since:

$$0.0741 > 0.0556$$

the model predicts:

Spam

Normalized probabilities:

Total:

$$0.0741 + 0.0556 = 0.1297$$

Spam:

$$P(\text{Spam}|X) = \frac{0.0741}{0.1297} \approx 0.5714$$

Not Spam:

$$P(\text{NotSpam}|X) = \frac{0.0556}{0.1297} \approx 0.4286$$

Therefore:

$$P(\text{Spam}|X) = 57.14\%$$

$$P(\text{NotSpam}|X) = 42.86\%$$

Final classification:

The email is classified as SPAM

4. Compare Results Using WEKA

In **WEKA**, the same dataset can be entered into an ARFF file and the **NaïveBayes** classifier can be selected.

Expected WEKA result

For:

Offer = Yes, Free = No, Length = Short

the Naïve Bayes classifier should classify the email as:

Spam

Our manual calculation also gives:

Spam = 57.14%

Not Spam = 42.86%

Therefore, the manual result and WEKA classification should agree, assuming the same dataset and settings are used.

Note: Because this is a very small dataset, WEKA's exact displayed probabilities can differ slightly depending on its probability estimation/smoothing settings. The predicted class should remain **Spam** for this example.

5. Independence Assumption in Naïve Bayes

Naïve Bayes assumes that all features are **conditionally independent given the class**.

For this problem, it assumes that:

- Offer
- Free
- Length

are independent of each other once we know whether the email is **Spam** or **Not Spam**.

For example:

is approximated as:

This assumption makes Naïve Bayes **simple and fast**.

Limitation

In real emails, these features may not actually be independent.

For example:

- Emails containing "**Free**" may frequently also contain "**Offer**".
- Email length may be related to the words used in the email.
- Spam emails may have combinations of features rather than independent features.

Therefore, the independence assumption may not always be completely realistic. However, **Naïve Bayes can still perform well even when the independence assumption is not perfectly true**.

Final Answers:

Question	Answer
P(Spam)	0.5 (50%)
P(Not Spam)	0.5 (50%)
TP/Conditional probabilities	Calculated above
Spam score for given email	0.0741
Not Spam score	0.0556
Normalized Spam probability	57.14%
Normalized Not Spam probability	42.86%
Final classification	SPAM
WEKA result	SPAM
Independence assumption	Features are assumed conditionally independent given the class

